3.3 Cramer's Rule

Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $Ax=b$ is affected by changes in the entries of b.

For any $n \times n$ matrix A and any b in $\mathbb{R}^n$, let $A_i(b)$ be the matrix obtained from A by replacing column i by the vector b.

$$A_i(b) = \begin{bmatrix} a_1 & \cdots & \underset{\uparrow}{b} & \cdots & a_n \\ & & \text{col } i & & \end{bmatrix}$$

Theorem: Let A be an invertible matrix. For any b in $\mathbb{R}^n$, the unique x of $Ax=b$ has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \quad i = 1, 2, \ldots, n.$$

Proof: Denote the columns of A by $a_1, \ldots, a_n$ and the columns of the $n \times n$ identity matrix I by $e_1, \ldots, e_n$. If $Ax=b$, the definition of matrix multiplication shows that

$$A \cdot I_i(x) = A\begin{bmatrix} e_1 & \cdots & x & \cdots & e_n \end{bmatrix}$$

$$= \begin{bmatrix} Ae_1 & \cdots & Ax & \cdots & Ae_n \end{bmatrix}$$

$A_{n\times n} \cdot e_{1\,n\times 1} = (A \cdot e_1)_{n\times 1}$

$$= [a_1 \ \cdots \ b \ \cdots \ a_n]$$

$$= A_i(b)$$

By the multiplicative property of determinants

$$\det(A) \cdot \det(I_i(x)) = \det A_i(b)$$

The second determinant on the left is simply x_i.

column i

$$I_i(b) = \begin{bmatrix} 1 & 0 & \cdots & x_1 & \cdots & 0 \\ 0 & 1 & & x_2 & & \vdots \\ \vdots & & & \vdots & & \\ & & & x_n & & 1 \end{bmatrix}$$

To see it make a cofactor expansion along the ith row.

Hence, $\det A \cdot x_i = \det A_i(b)$

because A is invertible and $\det A \neq 0$. ∎

So

$$\boxed{x_i = \frac{\det A_i(b)}{\det A}}$$

Ex: Use Cramer's rule to solve the system:

$$3x_1 - 2x_2 = 6$$
$$-5x_1 + 4x_2 = 8$$

Sol: View the system as $Ax = b$.

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix} \qquad A_1(b) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(b) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

$\det A = 2$. → the system has a unique solution.

By Cramer's rule $x_1 = \dfrac{\det A_1(b)}{\det A} = \dfrac{24+16}{2} = 20$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{24+30}{2} = 27$$

Ex: Consider the following system in which s is an unspecified parameter. Determine the values of s for which the system has a unique solution, and use Cramer's rule to describe the solution.

$$3s\,x_1 - 2x_2 = 4$$
$$-6\,x_1 + 5x_2 = 1$$

Sol: View the system as $Ax = b$.

Then $A = \begin{bmatrix} 3s & -2 \\ -6 & 5 \end{bmatrix}$ $A_1(b) = \begin{bmatrix} 4 & -2 \\ 1 & 5 \end{bmatrix}$, $A_2(b) = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}$

$\det A = 3s^2 - 12 = 3\cdot(s-2)(s+2)$

The system has a unique solution when $\det A \neq 0$
i.e. $s \neq \pm 2$

For such an s $\quad x_1 = \dfrac{\det A_1(b)}{\det A} = \dfrac{4s+2}{3(s+2)(s-2)}$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{3s+24}{3\cdot(s+2)(s-2)} = \frac{s+8}{(s+2)(s-2)}$$

Cramer's rule leads easily to a general formula for the inverse of an $n \times n$ matrix A.

The jth column of A^{-1} is a vector that satisfies

$$A \cdot x = e_j$$

$A \cdot A^{-1} = I$, let $X = A^{-1} \Rightarrow A \cdot X = I$

$\Rightarrow A x_j = e_j$ because $A \cdot [x_1 \; \cdots \; x_n] = [e_1 \; \cdots \; e_n]$

$\Rightarrow A x_j = e_j$

where e_j is the jth column of the identity matrix and ith entry of x is the (i,j)-entry of A^{-1}. By Cramer's rule

$$\{(i,j)\text{-entry of } A^{-1}\} = x_i = \frac{\det A_i(e_j)}{\det A} \qquad (2)$$

Recall that A_{ji} denotes the submatrix of A formed by deleting row j and column i. A cofactor expansion down column i of $A_i(e_j)$ shows that

$$\det A_i(e_j) = (-1)^{i+j} \cdot \det A_{ji} = C_{ji} \qquad (3)$$

where C_{ji} is a cofactor of A.

$A_i(e_j) = [a_1 \; \cdots \; e_j \; \cdots \; a_n]$ (column i)

row j $\longrightarrow$ $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$

By Eq. (2), the (i,j)-entry of A^{-1} is the cofactor C_{ji} divided by $\det A$.

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \qquad (4)$$

The matrix of cofactors on the right side of Eq. (4) is called the **adjugate of A**, denoted by **adj A**

Theorem: Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

Ex: Find inverse of $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$ using adj A.

Sol the nine cofactors are:

$C_{11} = +\begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2$ $\quad C_{21} = -\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3$ $\quad C_{31} = +\begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$

$C_{12} = -\begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14$ $\quad C_{22} = +\begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7$ $\quad C_{32} = -\begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$

$C_{13} = +\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = 4$ $\quad C_{23} = -\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1$ $\quad C_{33} = +\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$

The adjugate matrix is the transpose of the matrix of cofactors.

$$\operatorname{adj} A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

$A^{-1} \cdot A = \frac{1}{\det A} \operatorname{adj} A \cdot A$

$I \qquad\qquad 14 I$

$\Rightarrow \det A \cdot I = 14 \cdot I \Rightarrow \boxed{\det A = 14}$

We can compute $\det A$ directly, but the following computation provides a check on the calculations above and provides $\det A$.

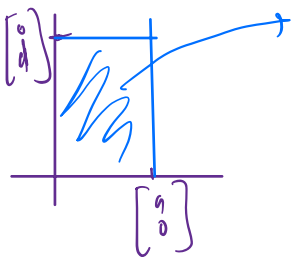
$$(\operatorname{adj} A) \cdot A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14 \cdot I$$

Since $(\operatorname{adj} A) A = 14 I$, theorem shows that $\det A = 14$ and

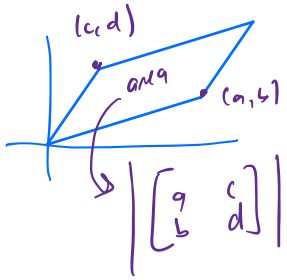
$$A^{-1} = \frac{1}{14} \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

Determinants as Area or Volume.

Theorem: If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.



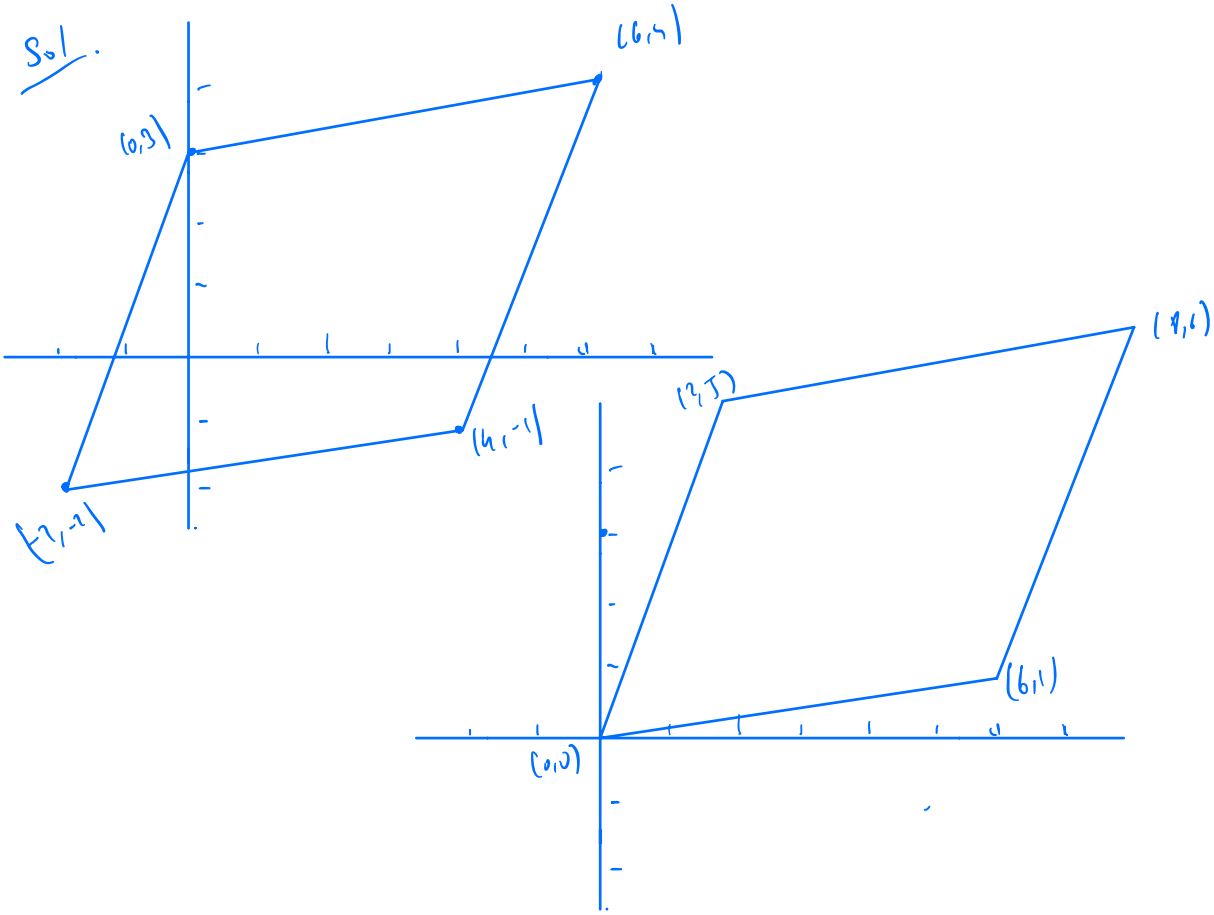
$$\left|\det\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}\right| = |ad| = \text{area of rectangle.}$$



$$\left|\det\begin{bmatrix} a & c \\ b & d \end{bmatrix}\right|$$

Ex: Calculate the area of the parallelogram determined by the points $(-2,-2)$, $(0,3)$, $(4,-1)$ and $(6,4)$

Sol.



Translate the parallelogram to one having the origin as a vertex.

For example subtract the vertex $(-2, -2)$ from each of the four vertices. The new parallelogram has the same area, and its vertices are $(0,0)$, $(6,1)$, $(2,5)$, $(8,6)$.

The parallelogram is determined by the columns of $A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}$

since $\det A = -28$ the area is 28.